

SAGNAW COUNTY AP, MI, USA

WMO: 722125

Lat: 43.433N

Lon: 83.867W

Elev: 183

StdP: 99.15

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 04829

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.5	-14.9	-22.0	0.5	-16.8	-18.8	0.7	-13.8	12.1	-3.4	11.0	-2.8	3.0	270	0.548

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	32.0	22.2	30.2	21.5	27.9	20.4	24.0	29.3	23.0	27.8	22.1	26.5	4.8	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.4	17.5	26.6	21.4	16.4	25.5	21.0	16.0	25.1	72.8	28.6	68.9	27.7	65.8	26.5	28.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.8	9.2	8.3	-20.3	33.9	2.9	1.5	-22.4	35.0	-24.1	35.9	-25.7	36.8	-27.8	37.9		
			-20.7	25.8	2.8	1.1	-22.7	26.6	-24.4	27.2	-25.9	27.8	-28.0	28.6		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.0	-4.1	-4.2	1.2	7.9	14.6	19.6	21.7	20.4	16.9	10.5	4.0	-1.3	(6)
	DBStd	10.42	5.83	5.72	6.02	4.85	4.93	3.70	3.28	3.02	4.10	4.69	4.79	4.80	(7)
	HDD10.0	1815	438	397	281	96	11	0	0	0	1	52	188	352	(8)
	HDD18.3	3721	696	630	531	313	140	27	7	14	76	248	429	610	(9)
	CDD10.0	1453	1	0	9	34	153	288	362	323	207	67	9	0	(10)
	CDD18.3	316	0	0	1	1	24	65	111	78	32	4	0	0	(11)
	CDH23.3	2864	0	0	7	21	262	605	1050	607	272	41	0	0	(12)
	CDH26.7	809	0	0	2	1	69	179	342	143	66	8	0	0	(13)
(14) Wind	WSAvg	3.7	4.2	4.3	4.1	4.4	3.8	3.2	2.8	2.6	2.9	3.7	4.0	4.1	(14)
(15) Precipitation	PrecAvg	800	49	43	49	77	89	87	81	83	72	73	66	50	(15)
	PrecMax	1016	92	117	121	171	198	185	125	130	145	143	139	111	(16)
	PrecMin	651	6	7	7	30	36	30	18	34	11	24	17	15	(17)
	PrecStd	102	24	27	26	37	40	36	29	27	37	33	37	22	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.3	13.0	22.9	26.2	31.2	32.8	34.0	32.4	31.3	27.5	20.1	13.8	(19)
		MCWB	11.4	9.6	16.2	16.2	20.8	22.2	23.7	22.2	21.3	20.4	14.0	11.7	(20)
	2%	DB	8.8	8.8	17.3	22.7	28.0	31.1	32.3	30.1	28.0	23.8	16.2	9.7	(21)
		MCWB	7.5	6.6	12.3	13.9	18.9	21.7	22.9	22.2	20.6	17.0	11.7	8.2	(22)
	5%	DB	5.2	5.1	13.0	20.0	26.4	28.7	30.3	27.8	26.3	21.2	13.7	7.1	(23)
		MCWB	3.4	2.9	9.3	12.9	17.7	21.2	21.9	20.8	19.2	16.6	10.1	5.7	(24)
	10%	DB	2.7	2.7	10.2	17.1	23.7	27.1	27.9	27.0	24.2	18.0	12.0	4.8	(25)
		MCWB	1.8	1.4	6.9	10.9	16.3	19.1	20.6	20.1	18.3	14.4	9.3	3.2	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.6	10.9	17.1	18.0	22.4	24.4	25.6	25.0	23.7	21.6	15.2	12.6	(27)
		MCDB	12.1	13.1	23.9	23.0	28.0	30.8	31.5	29.1	28.3	26.6	17.7	13.6	(28)
	2%	WB	7.7	6.8	13.3	15.4	20.8	22.8	24.4	23.3	22.2	19.3	12.5	8.1	(29)
		MCDB	8.4	9.3	16.2	20.6	25.0	28.5	29.9	27.8	26.3	22.2	15.3	8.9	(30)
	5%	WB	3.9	3.3	10.4	13.6	19.4	21.8	23.3	22.3	20.9	17.2	10.8	5.7	(31)
		MCDB	5.4	4.9	13.4	18.9	24.6	26.4	28.0	26.5	24.1	19.5	13.6	7.0	(32)
	10%	WB	2.0	1.5	6.8	11.5	17.9	20.7	22.3	21.4	19.7	15.0	8.7	3.0	(33)
		MCDB	2.8	2.9	9.7	16.6	22.6	25.1	26.6	24.9	23.5	18.3	11.1	4.2	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	6.8	7.7	8.9	11.1	11.7	11.2	11.4	10.9	11.6	9.7	7.5	5.6	(35)
		MCDBR	8.6	8.8	13.7	15.8	13.9	12.7	13.0	11.9	13.2	13.7	11.4	8.3	(36)
	5% WB	MCWBR	7.7	7.2	9.5	9.1	7.0	6.3	5.9	5.8	6.7	8.1	8.1	7.3	(37)
		MCWBR	8.4	8.7	12.9	14.0	12.4	10.8	10.9	9.9	11.2	11.6	10.4	8.3	(38)
(40) Clear-Sky Solar Irradiance	taub		0.292	0.314	0.335	0.369	0.394	0.422	0.411	0.403	0.376	0.346	0.312	0.288	(40)
		taud	2.380	2.321	2.375	2.306	2.293	2.245	2.289	2.302	2.374	2.449	2.490	2.442	(41)
	Ebn at Noon		839	878	905	895	879	852	857	851	850	834	814	814	(42)
		Edh at Noon	78	98	106	123	129	136	128	123	106	86	70	67	(43)
(44) All-Sky Solar Radiation	RadAvg		1.41	2.29	3.57	4.58	5.38	6.05	6.08	5.19	4.14	2.60	1.58	1.12	(44)
	RadStd		0.14	0.29	0.38	0.39	0.47	0.34	0.29	0.22	0.40	0.25	0.20	0.10	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (58 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page